

# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$d = -\frac{\{\phi_3^G, \Phi_1\}}{\{\phi_2^G, \Phi_1\}}, \quad v = -\frac{\{\phi_3^G, \Phi_6\}}{\{\Phi_7, \Phi_6\}}, \quad u = -\frac{\{\phi_3^G + v\phi_1^{(1)}, \phi_4^G\}}{\{\phi_2^{(1)}, \phi_4^G\}}.$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\psi_{(2)i}^\gamma(\mathbf{r}) = -f^{\alpha\beta b} f^{b\delta\gamma} \mathcal{X}^\alpha(\mathbf{r}) \mathcal{Q}_{(2)i}^\beta(\mathbf{r}) \mathcal{X}^\delta(\mathbf{r}) = -f^{\alpha\beta b} f^{b\delta\gamma} \mathcal{X}^\alpha(\mathbf{r}) [a_i^\beta(\mathbf{r}) + \frac{2}{3}x_i^\beta(\mathbf{r})] \mathcal{X}^\delta(\mathbf{r}),$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$(\Pi(t_\nu) - \Pi(t_\lambda)) C_{\lambda\mu}^\nu = \sum_{\tilde{\lambda}/d'/\lambda} \beta^{|\tilde{\lambda}/d/\lambda|} C_{\tilde{\lambda}\mu}^\nu - \sum_{\nu/d''/\tilde{\nu}} \beta^{|\nu/d''/\tilde{\nu}|} \Pi(t_{\tilde{\nu}}) C_{\lambda\mu}^{\tilde{\nu}},$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\Phi_{\Xi}(\zeta, \zeta^*) = \langle \exp(\zeta + \zeta^*) \mid \exp(\varsigma + \tau^*) \rangle e^{-\langle \tau^* | \varsigma \rangle} = e^{\langle \zeta | \tau^* \rangle + \langle \zeta^* | \varsigma \rangle - \langle \tau^* | \varsigma \rangle}$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\int_0^{2\pi} d\sigma X^i(\sigma) \hat{P}_i(\sigma) = x_0^i \hat{p}_0^i + \frac{g_{ij}}{2\sqrt{\alpha'}} \sum_{n=1}^{\infty} \left\{ \left( \chi_n^i \alpha_{-n}^j + \bar{\chi}_n^i \tilde{\alpha}_{-n}^j \right) + \left( \bar{\chi}_n^i \alpha_n^j + \chi_n^i \tilde{\alpha}_n^j \right) \right\}.$$